

Vpc challenges 4

1) Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.

—> create multiple vpc's and give different cidr ranges for each vpc's

The screenshot shows the AWS VPC dashboard. On the left is a navigation menu with options like 'Virtual private cloud', 'Subnets', 'Route tables', etc. The main area displays 'Your VPCs' with a table listing three VPCs:

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR
default	vpc-07968786753c2ea72	Available	—	—	Off	172.31.0.0/16	—
myvpc1	vpc-0bd93d506aa0a1f36	Available	—	—	Off	10.0.0.0/28	—
myvpc2	vpc-0250eb93b24951001	Available	—	—	Off	10.0.1.0/24	—

—> cidr 172.31.0.0/16 vpc default

—> cidr 10.0.0.0/28 vpc 1

—> cidr 10.0.1.0/28 vpc 2

After creating the vpc's create three subnets for three vpc's

—> attach each subnet too each vpc

—> create the transit gateway were wet are going to attach all the vpc's inside it

The screenshot shows the AWS Transit Gateway console. A green notification bar at the top states: 'You successfully created tgw-024db055556f9d5c9 / test-tg.' Below this, a table lists the transit gateway:

Name	Transit gateway ID	Owner ID	State
test-tg	tgw-024db055556f9d5c9	207662791773	Pending

—> go to transit gateway attachment

VPC > Transit gateway attachments

You successfully created VPC attachment tgw-attach-087ee0401372dc586 / default-tga.

Introducing new feature: Metering Policy for Transit Gateway (TGW)
 Previously, transit gateway data usage was metered solely to the source attachment owner. Now you can create custom metering policies to allocate data usage to source, destination, or central TGW accounts. [Learn more](#)

0 0 0 0 2 0

Transit gateway attachments (1) Info

Find transit gateway attachment by attribute or tag

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID	Association route
default-tga	tgw-attach-087ee0401372dc586	tgw-024db055556f9d5c9	Pending	VPC	vpc-07968786753c2ea72	-

- > after creating transit gateway go to transit gateway attachments
- > keep adding all the vpc's to the transit gateway
- > attached the default vpc to transit gateway

Network Firewall rule groups
 TLS inspection configurations
 Network Firewall resource groups
 VPC endpoint associations

Virtual private network (VPN)
 Customer gateways
 Virtual private gateways
 Site-to-Site VPN
 Concentrators [New](#)
 Site-to-Site VPN connections
 Client VPN endpoints

AWS Verified Access
 Verified Access instances
 Verified Access trust providers
 Verified Access groups
 Verified Access endpoints

You successfully created VPC attachment tgw-attach-0e3cf8ff4fe7c7462 / vpc2-tga.

Introducing new feature: Metering Policy for Transit Gateway (TGW)
 Previously, transit gateway data usage was metered solely to the source attachment owner. Now you can create custom metering policies to allocate data usage to source, destination, or central TGW accounts. [Learn more](#)

0 0 0 0 2 0

Transit gateway attachments (3) Info

Find transit gateway attachment by attribute or tag

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID	Association route
vpc1-tga	tgw-attach-03d11a4501cb12f07	tgw-024db055556f9d5c9	Available	VPC	vpc-0bd93d506aa0a1f36	tgw-rtb-0b27ec65
default-tga	tgw-attach-087ee0401372dc586	tgw-024db055556f9d5c9	Available	VPC	vpc-07968786753c2ea72	tgw-rtb-0b27ec65
vpc2-tga	tgw-attach-0e3cf8ff4fe7c7462	tgw-024db055556f9d5c9	Pending	VPC	vpc-06efa456995169bce	-

Select a transit gateway attachment

—> in the same way attach all vpc 's attach to the transit gateway

—> now after the transit gateway is created it will default create a route table for that transit gateway

VPC > Transit gateway route tables

Network Firewall rule groups
 TLS inspection configurations
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You successfully created VPC attachment tgw-attach-0e3cf8ff4fe7c7462 / vpc2-tga.

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 Previously, transit gateway data usage was metered solely to the source attachment owner. Now you can create custom metering policies to allocate data usage to source, destination, or central TGW accounts. [Learn more](#)

0 0 0 0 2 0

Transit gateway route tables (1) Info

Find transit gateway route table by attribute or tag

Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table	Default propagation
tgw-rtb-0b27ec658dddf4aeb	tgw-rtb-0b27ec658dddf4aeb	tgw-024db055556f9d5c9	Available	Yes	Yes

Transit gateway route tables (1/1) [Info](#)

Find transit gateway route table by attribute or tag

[Actions](#) [Create transit gateway route table](#)

☒	Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table	Default propagati...
☒	tgw-rtb-0b27ec658dddf4aeb	tgw-rtb-0b27ec658dddf4aeb	tgw-024db055556f9d5c9	Available	Yes	Yes

Transit gateway route tables: tgw-rtb-0b27ec658dddf4aeb

[Details](#) [Associations](#) [Propagations](#) [Prefix list references](#) [Routes](#) [Tags](#)

Associations (3) [Info](#)

Find association by attribute or tag

[Delete association](#) [Create association](#)

☐	Attachment ID	Resource type	Resource ID	State
☐	tgw-attach-0e3cf8ff4fe7c7462	VPC	vpc-06efa456995169bce	Associated
☐	tgw-attach-03d11a4501cb12f07	VPC	vpc-0bd93d506aa0a1f36	Associated
☐	tgw-attach-087ee0401372dc586	VPC	vpc-07968786753c2ea72	Associated

—> we have to attach this transit gateway to all vpc's subnet' s route

—> we can see this route is associated only to vpc's which we attached to transit gateway

—> now launch the instances with each vpc's and subnets which we created

EC2 > **Instances**

Instances (3) [Info](#)

Find Instance by attribute or tag (case-sensitive)

[Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Last updated less than a minute ago

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv
☐	default-vpc-ec2	i-04e774a1a95545bfc	Running	t3.micro	Initializing	View alarms +	us-east-1a	ec2-100-27-1-190.com...	100.27.1.1
☐	vpc2-ec2	i-0e453c9bf29ca61b9	Running	t3.micro	—	View alarms +	us-east-1a	—	3.238.89.1
☐	vpc1-ec2	i-0e30691f8e6bb2754	Running	t3.micro	Initializing	View alarms +	us-east-1f	—	98.93.27.1

—> now edit the route table off all subnets add the transit gateway route to it

—> for now we are inside the default vpc's route and we are adding vpc1 and vpc2 cidr ranges with the transit gateway with we created

Destination	Target	Status	Propagated	Route Origin
172.31.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute
10.0.0.0/28	igw-05d93d6a24c45086c	Active	No	CreateRoute
10.0.1.0/28	tgw-024db055556f9d5c9	Active	No	CreateRoute

—> we have attached the vpc1 vpc2 transit gateway to it in default vpc

Updated routes for rtb-0b5d29d80613137e7 / default-rt successfully

rtb-0b5d29d80613137e7 / default-rt

Details

Route table ID: rtb-0b5d29d80613137e7

VPC: vpc-07968786753c2ea72 | default

Main: Yes

Owner ID: 207662791773

Routes (4)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-05d93d6a24c45086c	Active	No	Create Route
10.0.0.0/28	tgw-024db055556f9d5c9	Active	No	Create Route
10.0.1.0/28	tgw-024db055556f9d5c9	Active	No	Create Route
172.31.0.0/16	local	Active	No	Create Route Table

—> this are the vpc1 vpc2 vpc default route tables which we. Created and going to attach the transit gateway to it

VPC > Route tables

PC dashboard <

NS Global View

Filter by VPC

Virtual private cloud

Subnets

Route tables

Internet gateways

Press-only internet gateways

Private gateways

EC2 option sets

Static IPs

Route tables (3) Info

Last updated less than a minute ago

Actions Create route table

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	default-rt	rtb-0b5d29d80613137e7	-	-	Yes	vpc-07968786753c2ea72 defa...	207662791773
☐	vpc2-rt	rtb-0d9fe83fd1fa635ae	-	-	Yes	vpc-06efa456995169bce vpc2	207662791773
☐	vpc1-rt	rtb-0f764cc462ca4e7a6	-	-	Yes	vpc-0bd93d506aa0a1f36 vpc1	207662791773

—> similarly add all the transit gateway to all vpc 's route

Vpc2 route

VPC > Route tables > rtb-0d9fe83fd1fa635ae > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.1.0/28	local	Active	No	CreateRouteTable
10.0.0.0/28	local	-	No	CreateRoute
172.31.0.0/16	Transit Gateway	-	No	CreateRoute
172.31.0.0/16	tgw-024db055556f9d5c9	-	No	CreateRoute

Add route

Cancel Preview Save changes

—> vpc1 route

VPC > Route tables > rtb-0f764cc462ca4e7a6 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/28	local	Active	No	CreateRouteTable
172.31.0.0/16	local	-	No	CreateRoute
172.31.0.0/16	Transit Gateway	-	No	CreateRoute
10.0.1.0/28	tgw-024db055556f9d5c9	-	No	CreateRoute

Add route

Cancel Preview Save changes

—> now create internet gateways to all vpc routes

The screenshot shows the AWS VPC console's 'Internet gateways' page. A green notification banner at the top states: 'Internet gateway igw-00339370171070665 successfully attached to vpc-06efa456995169bce'. Below this, the 'Internet gateways (3)' section displays a table with the following data:

Name	Internet gateway ID	State	VPC ID	Owner
default-vpc	igw-05d93d6a24c45086c	Attached	vpc-07968786753c2ea72 default	207662791773
vpc1-igw	igw-011c66301b5057b6d	Attached	vpc-0bd93d506aa0a1f36 vpc1	207662791773
vpc2-igw	igw-00339370171070665	Attached	vpc-06efa456995169bce vpc2	207662791773

—> we created vpc1-igw for vpc1
And vpc2-igw for vpc2

—> now add this gateway to all the vpc's route

The screenshot shows the 'Edit routes' page for route table 'rtb-0b5d29d80613137e7'. It displays a table of routes with the following data:

Destination	Target	Status	Propagated	Route Origin
172.31.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute
10.0.0.0/28	Transit Gateway	-	No	CreateRoute
10.0.1.0/28	Transit Gateway	-	No	CreateRoute

Buttons for 'Add route', 'Cancel', 'Preview', and 'Save changes' are visible at the bottom.

The screenshot shows the 'Details' page for route table 'rtb-0b5d29d80613137e7'. A green notification banner at the top states: 'Updated routes for rtb-0b5d29d80613137e7 / default-rt successfully'. The page includes sections for 'Details', 'Main', 'Explicit subnet associations', and 'Edge associations'. Below these, the 'Routes (4)' section displays a table with the following data:

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-05d93d6a24c45086c	Active	No	Create Route
10.0.0.0/28	tgw-024db055556f9d5c9	Active	No	Create Route
10.0.1.0/28	tgw-024db055556f9d5c9	Active	No	Create Route
172.31.0.0/16	local	Active	No	Create Route Table

—> again go to routes and add the internet gateways to all routes

VPC > Route tables

PC dashboard <

NS Global View ⓘ

Filter by VPC ▾

Virtual private cloud

- Subnets
- Route tables
- Internet gateways
- Press-only internet gateways
- Private gateways
- EC2 option sets
- Elastic IPs

Route tables (3) Info

Find route tables by attribute or tag

Last updated less than a minute ago Actions Create route table

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	default-rt	rtb-0b5d29d80613137e7	–	–	Yes	vpc-07968786753c2ea72 defa...	207662791773
☐	vpc2-rt	rtb-0d9fe83fd1fa635ae	–	–	Yes	vpc-06efa456995169bce vpc2	207662791773
☐	vpc1-rt	rtb-0f764cc462ca4e7a6	–	–	Yes	vpc-0bd93d506aa0a1f36 vpc1	207662791773

VPC > Route tables > rtb-0d9fe83fd1fa635ae > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.1.0/28	local	Active	No	CreateRouteTable
10.0.0.0/28	Transit Gateway	–	No	CreateRoute
172.31.0.0/16	Transit Gateway	–	No	CreateRoute

Add route

Cancel Preview Save changes

Security group inbound rule

—> in last we have to give all traffic to all instances

sg-0cb0dfb2177545b2a - default Actions

Details

Security group name default	Security group ID sg-0cb0dfb2177545b2a	Description default VPC security group	VPC ID vpc-0bd93d506aa0a1f36
Owner 207662791773	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules Outbound rules Sharing - new VPC associations - new Tags

Inbound rules (2) Manage tags Edit inbound rules

Search

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	–	sgr-0f8b6da6038c54f6c	IPv4	All traffic	All	All	0.0.0.0/0
☐	–	sgr-0a26ee420bc8d7620	–	All traffic	All	All	sg-0cb0dfb2177545b2a...

—> now go inside any instance and ping the other instance private ip

```

#_
~\_####_ Amazon Linux 2023
~~~\_#####\
~~~\_###|
~~~\_#/____ https://aws.amazon.com/linux/amazon-linux-2023
~~~V~' '->
~~~~
~~~~_. _/
~~~~/_/_/
~~~~/_m/'

[[ec2-user@ip-172-31-8-89 ~]$ ping 10.0.1.8
PING 10.0.1.8 (10.0.1.8) 56(84) bytes of data.
^C
--- 10.0.1.8 ping statistics ---
829 packets transmitted, 0 received, 100% packet loss, time 861097ms

[[ec2-user@ip-172-31-8-89 ~]$ ping 10.0.1.8
PING 10.0.1.8 (10.0.1.8) 56(84) bytes of data.
64 bytes from 10.0.1.8: icmp_seq=110 ttl=126 time=0.911 ms
64 bytes from 10.0.1.8: icmp_seq=111 ttl=126 time=0.416 ms
64 bytes from 10.0.1.8: icmp_seq=112 ttl=126 time=0.410 ms
64 bytes from 10.0.1.8: icmp_seq=113 ttl=126 time=0.555 ms
64 bytes from 10.0.1.8: icmp_seq=114 ttl=126 time=0.412 ms
64 bytes from 10.0.1.8: icmp_seq=115 ttl=126 time=0.427 ms
64 bytes from 10.0.1.8: icmp_seq=116 ttl=126 time=0.459 ms
64 bytes from 10.0.1.8: icmp_seq=117 ttl=126 time=0.423 ms
64 bytes from 10.0.1.8: icmp_seq=118 ttl=126 time=0.428 ms
64 bytes from 10.0.1.8: icmp_seq=119 ttl=126 time=0.454 ms
64 bytes from 10.0.1.8: icmp_seq=120 ttl=126 time=0.466 ms
64 bytes from 10.0.1.8: icmp_seq=121 ttl=126 time=0.412 ms

```

—> we can see through pic that my vpc default instance is success fully ping the other vpc instance

Warning: Permanently added '100.27.1.190' (ED25519) to the list of known hosts.

```

#_
~\ ##### Amazon Linux 2023
~~~\ #####
~~~\ #####|
~~~\ #/ _____ https://aws.amazon.com/linux/amazon-linux-2023
~~~~~V~' '~>
~~~~~
~~~~~\
~~~~~\
~~~~~\m/'

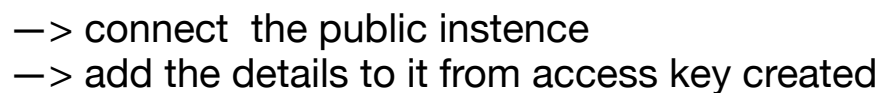
```

```
[ec2-user@ip-172-31-8-89 ~]$ ping 10.0.1.8
PING 10.0.1.8 (10.0.1.8) 56(84) bytes of data.
^C
--- 10.0.1.8 ping statistics ---
829 packets transmitted, 0 received, 100% packet loss, time 861097ms
```

```
[ec2-user@ip-172-31-8-89 ~]$ ping 10.0.1.8
PING 10.0.1.8 (10.0.1.8) 56(84) bytes of data.
64 bytes from 10.0.1.8: icmp_seq=110 ttl=126 time=0.911 ms
64 bytes from 10.0.1.8: icmp_seq=111 ttl=126 time=0.416 ms
64 bytes from 10.0.1.8: icmp_seq=112 ttl=126 time=0.410 ms
64 bytes from 10.0.1.8: icmp_seq=113 ttl=126 time=0.555 ms
64 bytes from 10.0.1.8: icmp_seq=114 ttl=126 time=0.412 ms
64 bytes from 10.0.1.8: icmp_seq=115 ttl=126 time=0.427 ms
64 bytes from 10.0.1.8: icmp_seq=116 ttl=126 time=0.459 ms
64 bytes from 10.0.1.8: icmp_seq=117 ttl=126 time=0.423 ms
64 bytes from 10.0.1.8: icmp_seq=118 ttl=126 time=0.428 ms
64 bytes from 10.0.1.8: icmp_seq=119 ttl=126 time=0.454 ms
64 bytes from 10.0.1.8: icmp_seq=120 ttl=126 time=0.466 ms
64 bytes from 10.0.1.8: icmp_seq=121 ttl=126 time=0.412 ms
64 bytes from 10.0.1.8: icmp_seq=122 ttl=126 time=0.463 ms
64 bytes from 10.0.1.8: icmp_seq=123 ttl=126 time=0.412 ms
64 bytes from 10.0.1.8: icmp_seq=124 ttl=126 time=0.430 ms
64 bytes from 10.0.1.8: icmp_seq=125 ttl=126 time=0.426 ms
AC
--- 10.0.1.8 ping statistics ---
125 packets transmitted, 16 received, 87.2% packet loss, time 128995ms
rtt min/avg/max/mdev = 0.410/0.469/0.911/0.119 ms
```

```
[ec2-user@ip-172-31-8-89 ~]$ ping 10.0.0.14
PING 10.0.0.14 (10.0.0.14) 56(84) bytes of data.
64 bytes from 10.0.0.14: icmp_seq=1 ttl=126 time=2.44 ms
64 bytes from 10.0.0.14: icmp_seq=2 ttl=126 time=1.02 ms
64 bytes from 10.0.0.14: icmp_seq=3 ttl=126 time=0.669 ms
64 bytes from 10.0.0.14: icmp_seq=4 ttl=126 time=0.761 ms
64 bytes from 10.0.0.14: icmp_seq=5 ttl=126 time=0.730 ms
64 bytes from 10.0.0.14: icmp_seq=6 ttl=126 time=0.729 ms
64 bytes from 10.0.0.14: icmp_seq=7 ttl=126 time=0.665 ms
64 bytes from 10.0.0.14: icmp_seq=8 ttl=126 time=1.05 ms
64 bytes from 10.0.0.14: icmp_seq=9 ttl=126 time=0.794 ms
64 bytes from 10.0.0.14: icmp_seq=10 ttl=126 time=0.650 ms
64 bytes from 10.0.0.14: icmp_seq=11 ttl=126 time=0.649 ms
64 bytes from 10.0.0.14: icmp_seq=12 ttl=126 time=0.679 ms
64 bytes from 10.0.0.14: icmp_seq=13 ttl=126 time=0.659 ms
64 bytes from 10.0.0.14: icmp_seq=14 ttl=126 time=0.682 ms
64 bytes from 10.0.0.14: icmp_seq=15 ttl=126 time=0.702 ms
```

- > launch two instances one with public subnet and one with private subnet
- > create a access key for aws configuring



- Aws s3 ls
- we see we are able to read the s3 services

—> now we have to give ssh -I <that file which we added pem> ec2-user@<private instance private ip >

```
-----END RSA PRIVATE KEY-----
waseemakram@waseem downloads % ssh -i "waseem.pem" ec2-user@ec2-56-228-12-233.eu-north-1.compute.amazonaws.com

#_
~\ ####_      Amazon Linux 2023
~~ \#####\
~~  \###|
~~   \#/  ____ https://aws.amazon.com/linux/amazon-linux-2023
~~    V~' '->
~~~~
~~~~
~m/'

Last login: Tue Nov 25 13:26:14 2025 from 14.192.14.62
[ec2-user@ip-172-31-21-156 ~]$ vi test.pem
[ec2-user@ip-172-31-21-156 ~]$ vi test.pem
[ec2-user@ip-172-31-21-156 ~]$ chmod 400 test.pem
[ec2-user@ip-172-31-21-156 ~]$ ssh -i test.pem ec2-user@172.31.40.194
The authenticity of host '172.31.40.194 (172.31.40.194)' can't be established.
ED25519 key fingerprint is SHA256:P+tYMVZ/QHVLIdX70wKcWftEcStTW+48L4P7wm9YwVI.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.40.194' (ED25519) to the list of known hosts.

#_
~\ ####_      Amazon Linux 2023
~~ \#####\
~~  \###|
~~   \#/  ____ https://aws.amazon.com/linux/amazon-linux-2023
~~    V~' '->
~~~~
~~~~
~m/'

[ec2-user@ip-172-31-40-194 ~]$ aws s3 ls

Unable to locate credentials. You can configure credentials by running "aws configure".
[ec2-user@ip-172-31-40-194 ~]$ aws configure
AWS Access Key ID [None]: AKIATAWNKXR07YXDDJAL
AWS Secret Access Key [None]: wkBAeW3Uu2kKI2yrUaL2zXHeLYa+LV1JsBxYJDu6
Default region name [None]: eu-north-1
Default output format [None]: json
[ec2-user@ip-172-31-40-194 ~]$ ls -a
.  ..  .aws  .bash_logout  .bash_profile  .bashrc  .ssh
[ec2-user@ip-172-31-40-194 ~]$ aws s3 ls
```

—> and give same aws configuration for private instance also
—> but in private instance we are unable to find the services like s3 because we don't have internet access like public instance

—> for over come this we are adding endpoint to the private instance
—> vpc endpoint is use for communicat with aws service without internet connectivity

—> create a endpoint

VPC > Endpoints > Create endpoint

Create endpoint [Info](#)

Create the type of VPC endpoint that supports the service, service network or resource to which you want to connect.

Endpoint settings

Specify a name and select the type of endpoint.

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify. Tags help you find and manage your endpoint.

s3-endpoint

Type [Info](#)

Select a category

☒ AWS services

Connect to services provided by Amazon with an Interface endpoint, or a Gateway endpoint

☐ PrivateLink Ready partner services

Connect to SaaS services which have AWS Service Ready designation with an Interface endpoint. Uses AWS PrivateLink

☐ AWS Marketplace

Connect to SaaS services with an Interface Endpoint

☐ EC2 Instance Connect Endpoint

An elastic network interface that allows you to connect to resources in a private subnet

☐ Resources

Connect to resources like Amazon Relational Database Services (RDS) with a Resource endpoint. Uses AWS PrivateLink

☐ Service network

Connect to VPC Lambda PrivateLink

☐ Endpoint services that use NLBs and GWLBs

Find services shared with you by service name. Connect to a Network LoadBalancer (NLB) service with an Interface endpoint or to a Gateway LoadBalancer (GWLB) service with a Gateway Load Balancer endpoint

Service Region

☒ Enable Cross Region endpoint [Info](#)

Connect to cross Region enabled services.

Europe (Stockholm) (eu-north-1)

—> give aws service as type

—> select s3 as service for which we want to communicate

Service Name = com.amazonaws.eu-north-1.s3 [Clear filters](#)

Service Name	Owner	Type
☒ com.amazonaws.eu-north-1.s3	amazon	Gateway
☐ com.amazonaws.eu-north-1.s3	amazon	Interface

Network settings

Select the VPC in which to create the endpoint

VPC
Create the VPC endpoint in the VPC in the same AWS Region from which you will access a resource.

vpc-01126d967b1bb73af

Additional settings

Route tables (1/2) [Info](#)

Search

Name	Route Table ID	Main	Associated Id
☐ -	rtb-0da8e9d61854221d2	Yes	2 subnets
☒ private-routh	rtb-0092524888a9180ad (private-routh)	No	subnet-05f27e3702839a75e (private-subnet)

When you use an endpoint, the source IP addresses from your instances in your affected subnets for accessing the AWS service in the same region will be private IP addresses, not public IP addresses. Existing connections from your affected subnets to the AWS service that use public IP addresses may be dropped. Ensure that you don't have critical tasks running when you create or modify an endpoint.

—> add vpc and add the private instance route

—> created the end point

Successfully created VPC endpoint
vpce-01332b206155aed1d

Endpoints (1/1) Info

Find endpoints by attribute or tag

VPC endpoint ID : [vpce-01332b206155aed1d](#) Clear filters

< 1 > ⚙

☒	Name	VPC endpoint ID	Endpoint type	Status	Service name
☒	s3-endpoint	vpce-01332b206155aed1d	Gateway	Available	com.amazonaws.eu-north-1.s3

vpce-01332b206155aed1d / s3-endpoint Actions

Details

Endpoint ID
[vpce-01332b206155aed1d](#)

VPC ID
[vpc-01126d967b1bb73af](#)

DNS record IP type
service-defined

Private DNS specified domains
-

Status
Available

Status message
-

IP address type
ipv4

Creation time
Tuesday, November 25, 2025 at 19:15:59 GMT+5:30

Service name
[com.amazonaws.eu-north-1.s3](#)

Service region
eu-north-1

Endpoint type
Gateway

Private DNS names enabled
No

Private DNS preference
-

Route tables

PolicyTags

Route tables (1)

Search

< 1 > ⚙

Name	Route Table ID	Main	Associated Id
private-routh	rtb-0092524888a9180ad (private-routh)	No	subnet-05f27e3702839a75e (private-subnet)

—> check the vpc 's details to check weather endpoint connected or not

R (Network border group)

Network Address Usage metrics
Disabled

Route 53 Resolver DNS Firewall rule groups
-

Owner ID
[207662791773](#)

Encryption control ID

Encryption control mode
-

Map

CIDRsFlow logsTagsIntegrations

Resource map Info

Show all details

Subnets (3)
Subnets within this VPC

eu-north-1a
A public-subnet

eu-north-1b
B private-subnet

eu-north-1c
C subnet-006e395d3b72e2ed7

Route tables (2)
Route network traffic to resources

private-routh

rtb-0da8e9d61854221d2

Network Connections (2)
Connections to other networks

igw-06827b286d7eba643

s3-endpoint

—> end point connected to private instance

Details [Info](#)

Route table ID
[rtb-0092524888a9180ad](#)

VPC
[vpc-01126d967b1bb73af](#)

Main
[No](#)

Owner ID
[207662791773](#)

Explicit subnet associations
[subnet-05f27e3702839a75e](#) / [private-subnet](#)

Edge associations
-

[Routes](#) | [Subnet associations](#) | [Edge associations](#) | [Route propagation](#) | [Tags](#)

Routes (2) [Both](#) [Edit routes](#)

Destination	Target	Status	Propagated	Route Origin
pl-c3aa4faa	vpce-01332b206155aed1d	Active	No	Create Route
172.31.0.0/16	local	Active	No	Create Route Table

```
ED25519 key fingerprint is SHA256: F6M1E2/qmVLE1x0mK6mEE5cM10E1H1MS1W1.
This key is not known by any other names
[Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.40.194' (ED25519) to the list of known hosts.

#_
~\ ####_      Amazon Linux 2023
~~ \#####\
~~  \###|
~~   \#/  ___  https://aws.amazon.com/linux/amazon-linux-2023
~~    V~' '->
~~~
~~~. _ . _/
~~~ /  /
~~~ /m/'

[ec2-user@ip-172-31-40-194 ~]$ aws s3 ls

Unable to locate credentials. You can configure credentials by running "aws configure".
[ec2-user@ip-172-31-40-194 ~]$ aws configure
AWS Access Key ID [None]: AKIATAWNKXR07YXDDJAL
AWS Secret Access Key [None]: wkBAeW3Uu2kKI2yrUaL2zXHeLYa+LV1JsBxYJDu6
Default region name [None]: eu-north-1
Default output format [None]: json
[ec2-user@ip-172-31-40-194 ~]$ ls -a
.  ..  .aws  .bash_logout  .bash_profile  .bashrc  .ssh
[ec2-user@ip-172-31-40-194 ~]$ aws s3 ls

Connect timeout on endpoint URL: "https://s3.eu-north-1.amazonaws.com/"
[ec2-user@ip-172-31-40-194 ~]$ aws s3 ls
2025-11-23 14:59:47 my-vpc-bucket-flowlog
[ec2-user@ip-172-31-40-194 ~]$
```

We are now able to communicate with aws s3 service without internet

